



DevOps Shack

50 Kubernetes Errors & Solutions

1. CrashLoopBackOff: Pod fails to start repeatedly.

- **Error Example:** Your pod status shows:

```
kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
myapp-12345-abcde	0/1	CrashLoopBackOff	10	5m

- **Cause:** This error typically indicates that the container crashes soon after starting.
- **Solution:** Check the logs to see why it's failing to start:

```
kubectl logs myapp-12345-abcde
```

- If there's an application error (e.g., a missing file or environment variable), correct the configuration or the container image itself. For instance:

```
# Deployment YAML Example
```

```
env:
```

```
- name: DATABASE_URL
```

```
  value: "jdbc:postgresql://mydb:5432/mydatabase"
```

2. ImagePullBackOff: Kubernetes can't pull the specified container image.

- **Error Example:**

```
kubectl describe pod myapp-12345-abcde
```

```
# Output
```

```
Failed to pull image "myregistry/myapp:latest": image not found
```

- **Cause:** This occurs if Kubernetes cannot find or access the specified container image.
- **Solution:**
 - Check the image name and tag for errors.
 - Ensure that your image is available in the container registry.
 - If using a private registry, make sure your Kubernetes cluster can access it with proper credentials, as shown below:

imagePullSecrets:

- name: myregistrykey

- Create a secret if necessary:

```
kubectl create secret docker-registry myregistrykey --docker-server=myregistry --docker-username=myuser --docker-password=mypass
```

3. ErrImagePull: Failure in pulling the image.

- **Solution:** This error is similar to ImagePullBackOff. Ensure:

- The image name is correct.
- The registry is accessible.
- Use a secure registry if necessary and validate permissions.
- An example command for pulling:

```
docker pull myregistry/myapp:latest
```

- You can also inspect node-level issues or check firewall settings if you're in a restricted network environment.
-

4. Pending Pods: Pods remain in pending state due to lack of resources or node unavailability.

- **Error Example:**

```
kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
myapp	0/1	Pending	0	1m

- **Solution:**

- Check the pod events with:

```
kubectl describe pod myapp
```

- If you see messages like "Insufficient CPU," then the cluster is out of resources. Either scale up your cluster, increase node resources, or decrease the pod's resource requests.

resources:

requests:

memory: "64Mi"

cpu: "250m"

5. Node NotReady: A node is not in a ready state, which prevents pods from running on it.

- **Error Example:**

kubectl get nodes

NAME	STATUS	ROLES	AGE	VERSION
worker-node	NotReady	<none>	30m	v1.20.0

- **Solution:**

- Describe the node to understand why it's not ready:

kubectl describe node worker-node

- Common causes include network issues, disk pressure, or kubelet failures. You may need to restart the kubelet or fix network configurations on the affected node.
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6. OOMKilled: The pod gets killed because it uses more memory than allocated.

- **Error Example:**

- You may find events similar to:

Last State: Terminated

Reason: OOMKilled

Exit Code: 137

- **Solution:**

- Increase the memory limit of the container:

resources:

limits:

memory: "128Mi"

requests:

memory: "64Mi"

- If your application is prone to memory spikes, you might want to optimize it or provide more resources.
-

7. Unauthorized: Authentication failure when trying to access resources.

- **Error Example:**

Error from server (Unauthorized): pods is forbidden: User

"system:serviceaccount:default:myserviceaccount" cannot list resource "pods" in API group "" in the namespace "default"

- **Solution:**

- Make sure the ServiceAccount is associated with the correct roles.

```
kind: RoleBinding
apiVersion: rbac.authorization.k8s.io/v1
metadata:
  name: read-pods
  namespace: default
subjects:
- kind: ServiceAccount
  name: myserviceaccount
  namespace: default
roleRef:
  kind: Role
  name: pod-reader
  apiGroup: rbac.authorization.k8s.io
```

8. Forbidden: Authorization issue where the user does not have permissions.

- **Error Example:**

```
Error from server (Forbidden): services is forbidden: User "user" cannot list resource "services" in API group ""
```

- **Solution:**

- This error generally occurs due to lack of access in RBAC. Check if the user or ServiceAccount has the correct role binding.
- Example:

```
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRoleBinding
metadata:
  name: my-clusterrolebinding
subjects:
- kind: User
  name: user
  apiGroup: rbac.authorization.k8s.io
roleRef:
  kind: ClusterRole
  name: cluster-admin
  apiGroup: rbac.authorization.k8s.io
```

9. Evicted Pods: Pods are removed from a node due to resource constraints.

- **Error Example:**

- When nodes are low on resources, pods may be evicted:

```
kubectl get pod myapp-12345-abcde
NAME             READY STATUS  RESTARTS  AGE
myapp-12345-abcde 0/1   Evicted  0         1m
```

- **Solution:**

- Either add resources to nodes or reduce resource requests in your workloads. Additionally, ensure resource limits are configured properly to avoid excessive resource consumption.
-

10. PVC Bound Issues: PersistentVolumeClaims (PVCs) are not bound to PersistentVolumes (PVs).

- **Error Example:**

- If a PVC status remains “Pending”:

```
kubectl get pvc
```

NAME	STATUS	VOLUME	CAPACITY	ACCESS MODES	STORAGECLASS	AGE
my-pvc	Pending	<none>	<none>	<none>	standard	5m

- **Solution:**

- Check that a matching PersistentVolume is available and meets the PVC requirements.

```
apiVersion: v1
kind: PersistentVolume
metadata:
  name: pv0001
spec:
  capacity:
    storage: 1Gi
  accessModes:
    - ReadWriteOnce
  persistentVolumeReclaimPolicy: Retain
  storageClassName: standard
  hostPath:
    path: "/mnt/data"
```

11. Service Not Accessible: Services are not reachable within or outside the cluster.

- **Error Example:**

- Trying to access a service at http://service-name:port but getting a timeout.

- **Solution:**

- Verify the service is correctly configured and check if pods are assigned to the service.
- Run:

```
kubectl describe service my-service
```

- Ensure service type (e.g., ClusterIP, NodePort, LoadBalancer) is correctly configured and check firewall settings if necessary.
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12. DNS Resolution Failures: Pods cannot resolve DNS names within the cluster.

- **Error Example:**
 - Pod logs may show errors like "Unable to resolve host: [hostname]."
- **Solution:**
 - Verify CoreDNS pods are running:

```
kubectl get pods -n kube-system -l k8s-app=kube-dns
```

- Restart CoreDNS if needed, and verify your pod's DNS settings:

```
dnsPolicy: ClusterFirst
```

13. Certificate Errors: TLS certificates are invalid or expired.

- **Error Example:**
 - Accessing a Kubernetes service with TLS and receiving "Certificate Expired" error.
- **Solution:**
 - Renew certificates with kubeadm certs renew (if using kubeadm). Update your TLS secret:

```
kubectl delete secret my-cert
```

```
kubectl create secret tls my-cert --cert=path/to/tls.crt --key=path/to/tls.key
```

14. API Server Unreachable: Cannot connect to the Kubernetes API server.

- **Error Example:**
 - kubectl commands fail with "Unable to connect to the server: connection refused."
- **Solution:**
 - Check the API server status with:

```
systemctl status kube-apiserver
```

- Ensure firewall rules allow access to port 6443 and review kube-apiserver logs for errors.
-

15. Scheduler Failures: Pods are not being scheduled.

- **Error Example:**
 - Pods stuck in Pending status without apparent resource issues.
 - **Solution:**
 - Check if the scheduler is running:
`kubectl get pods -n kube-system | grep kube-scheduler`
 - Review the scheduler logs and verify that pod anti-affinity rules aren't too restrictive.
-

16. Controller Manager Issues: Controllers aren't managing resources properly.

- **Error Example:**
 - Resources like Deployments or ReplicaSets not behaving as expected.
- **Solution:**
 - Check the controller manager pod's logs for errors and ensure it's running:

`kubectl logs -n kube-system kube-controller-manager-[pod-name]`

17. Network Plugin Errors: Issues with the network plugin can cause connectivity problems.

- **Error Example:**
 - Pods cannot communicate across nodes.
 - **Solution:**
 - Check if the network plugin (e.g., Calico, Flannel) pods are running and review their logs.
 - Inspect `kubectl get pods -n kube-system` to ensure the network plugin's pods are up.
-

18. Pod Stuck in Terminating State: Pod doesn't terminate after issuing delete command.

- **Solution:**
 - Force delete the pod:
`kubectl delete pod [pod-name] --grace-period=0 --force`
 - Investigate why the pod failed to terminate, such as open connections or finalizers.
-

19. ConfigMap Not Found: Pod references a missing ConfigMap.

- **Error Example:**

Warning FailedMount ... ConfigMap "myconfig" not found

- **Solution:**

- Create the missing ConfigMap:

```
kubectl create configmap myconfig --from-literal=key=value
```

20. Secret Not Found: Pod references a missing Secret.

- **Error Example:**

Warning FailedMount ... Secret "mysecret" not found

- **Solution:**

- Create or update the secret:

```
kubectl create secret generic mysecret --from-literal=username=admin --from-literal=password=pass
```

21. HPA Not Scaling: Horizontal Pod Autoscaler is not scaling as expected.

- **Error Example:**

- HPA status shows "Desired Replicas: 1" despite high load.

- **Solution:**

- Ensure the metrics server is running:

```
kubectl get deployment metrics-server -n kube-system
```

- Confirm HPA is targeting the correct metrics and configure thresholds if needed.
-

22. Ingress Not Working: Ingress does not route traffic as expected.

- **Error Example:**

- Ingress setup is complete, but external requests fail.

- **Solution:**

- Ensure an ingress controller (e.g., NGINX Ingress) is deployed.
- Check ingress resource configuration:

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
```



```
name: example-ingress
spec:
  rules:
  - host: example.com
    http:
      paths:
      - path: /
        pathType: Prefix
        backend:
          service:
            name: example-service
            port:
              number: 80
```

23. DaemonSet Pods Not Running: DaemonSet pods do not start on all nodes.

- **Error Example:**

```
kubectl get daemonsets -A
```

Shows fewer than expected pods.

- **Solution:**

- Review node taints/tolerations and add tolerations to the DaemonSet spec if needed.
-

24. Job Not Completing: Kubernetes Job fails to finish successfully.

- **Error Example:**

- Job remains in "Running" or "Failed" status.

- **Solution:**

- Check pod logs for errors and review job spec:

```
apiVersion: batch/v1
kind: Job
metadata:
  name: my-job
spec:
  backoffLimit: 3
  template:
    spec:
      containers:
      - name: my-container
        image: busybox
        command: ["echo", "Hello World"]
      restartPolicy: Never
```

25. PVC Pending: PersistentVolumeClaim remains in "Pending" status.

- **Error Example:**

- PVC status shows:

```
kubectl get pvc
```

NAME	STATUS	VOLUME	CAPACITY	ACCESS MODES	STORAGECLASS	AGE
my-pvc	Pending	<none>	<none>	<none>	standard	1h

- **Solution:**

- Ensure there's a matching PersistentVolume with the correct access modes and storage class.
-

26. Node Disk Pressure: Node has high disk usage, causing evictions.

- **Error Example:**

- Pods get evicted with DiskPressure status on the node.

- **Solution:**

- Free up space on the node or add additional storage, as shown:

```
df -h
```

27. Pod Affinity/Anti-Affinity Issues: Pods are unscheduled due to restrictive affinity rules.

- **Solution:**

- Ensure affinity rules are not too restrictive in the deployment YAML:

```
affinity:  
  podAntiAffinity:  
    requiredDuringSchedulingIgnoredDuringExecution:  
      - labelSelector:  
          matchExpressions:  
            - key: app  
              operator: In  
              values:  
                - myapp  
          topologyKey: "kubernetes.io/hostname"
```

28. ServiceAccount Not Found: Pods reference a missing ServiceAccount.

- **Solution:**

- Create or specify a valid ServiceAccount:

```
kubectl create serviceaccount myserviceaccount
```

29. Node NotSchedulable: Node is marked as unschedulable, preventing pods from being placed.

- **Error Example:**

```
kubectl get nodes
```

NAME	STATUS	ROLES	AGE	VERSION
worker-node	Ready,SchedulingDisabled	<none>	1h	v1.20.0

- **Solution:**
 - Enable scheduling on the node:

```
kubectl uncordon worker-node
```

30. Readiness Probe Failures: Containers fail readiness checks, causing them to stay in a “Not Ready” state.

- **Error Example:**
 - Describe the pod and see repeated readiness probe failures.
- **Solution:**
 - Adjust probe parameters to suit the application’s startup time:

```
readinessProbe:  
  httpGet:  
    path: /health  
    port: 8080  
  initialDelaySeconds: 10  
  periodSeconds: 5
```

31. Liveness Probe Failures: Containers fail liveness checks, resulting in restarts.

- **Solution:**
 - Similar to readiness probes, increase the delay and interval to allow the application more time to become live:

```
livenessProbe:  
  httpGet:  
    path: /live  
    port: 8080  
  initialDelaySeconds: 15  
  periodSeconds: 10
```

32. Namespace Not Found: Resource references a non-existent namespace.

- **Error Example:**

Error from server (NotFound): namespaces "test-namespace" not found

- **Solution:**

- Create the namespace before deploying resources:

```
kubectl create namespace test-namespace
```

33. ClusterRoleBinding Misconfiguration: Access issues due to incorrect ClusterRoleBinding setup.

- **Solution:**

- Check and configure the ClusterRoleBinding correctly:

```
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRoleBinding
metadata:
  name: example-binding
subjects:
  - kind: ServiceAccount
    name: example-sa
    namespace: default
roleRef:
  kind: ClusterRole
  name: view
  apiGroup: rbac.authorization.k8s.io
```

34. PVC Not Bound: PVC fails to bind to PV due to storage class mismatch.

- **Solution:**

- Ensure the storageClassName matches between PVC and PV.
-

35. Node Memory Pressure: Nodes experience high memory pressure, causing evictions.

- **Solution:**

- Check node memory usage with `kubectl top nodes` and consider scaling up nodes or reducing pod resource requests.
-

36. Service Endpoint Not Updated: Service does not update endpoints, resulting in unreachable services.

- **Solution:**

- Ensure that the selector matches pod labels:

```
selector:  
app: myapp
```

37. Endpoint Slices Issues: Endpoint slices are missing, causing network issues.

- **Solution:**
 - Reconfigure or manually create endpoint slices if needed. Ensure kube-proxy is running and check its logs.
-

38. DaemonSet Not Deploying on All Nodes: DaemonSet skips certain nodes.

- **Solution:**
 - Check taints on nodes and add tolerations to the DaemonSet:

```
tolerations:  
- key: "key"  
  operator: "Exists"
```

39. Finalizer Preventing Resource Deletion: Resource remains due to finalizers.

- **Solution:**
 - Remove finalizers to allow deletion:

```
kubectl patch resource resource-name -p '{"metadata":{"finalizers":[]}}' --type=merge
```

40. Ingress 404 Errors: Requests to Ingress return 404.

- **Solution:**
 - Ensure correct path definitions and verify that the Ingress controller is properly configured.
-

41. LoadBalancer IP Not Assigned: LoadBalancer service fails to get an external IP.

- **Solution:**
 - Verify cloud provider configuration or use a different service type like NodePort for testing.
-

42. HPA Targets Not Matching Metrics: HPA doesn't scale as it's not receiving target metrics.

- **Solution:**
 - Check the HPA target settings and ensure metrics are available via metrics server or Prometheus.
-

43. PersistentVolume Deleted but PVC Bound: PVC remains bound even though PV was deleted.

- **Solution:**
 - Manually unbind and delete PVC or recreate PV with the same name to bind back.
-

44. Helm Release Fails: Helm fails due to missing charts or resources.

- **Solution:**
 - Ensure Helm chart dependencies are installed using:

`helm dependency update mychart/`

45. API Version Deprecated: Using outdated API versions causes compatibility issues.

- **Solution:**
 - Update your YAML files to use the latest API versions (e.g., `apps/v1` instead of `extensions/v1beta1` for Deployments).
-

46. Namespace Resource Quota Exceeded: Deployments fail due to resource quota limits in a namespace.

- **Solution:**
 - Check quota with:

`kubectl describe quota -n [namespace]`

- Adjust resources or increase the quota if needed.
-

47. Cannot Attach Volume to Multiple Pods: Persistent volumes with RWO access can't be shared across pods.

- **Solution:**
 - Switch to a storage class supporting ReadWriteMany (RWX), like NFS, for shared access.
-

48. CPU Throttling: Containers experience high CPU throttling.

- **Solution:**
 - Increase the CPU limit or optimize application code to avoid exceeding CPU quotas:

```
resources:  
  limits:  
    cpu: "500m"  
  requests:  
    cpu: "250m"
```

49. Pods Evicted Due to Overcommit: Overcommitted resources lead to evictions.

- **Solution:**
 - Allocate resources more conservatively or scale cluster resources accordingly.
-

50. PodSecurityPolicy Issues: Pods fail to start due to restrictive PodSecurityPolicy.

- **Solution:**
 - Adjust the PodSecurityPolicy to allow necessary permissions:

```
apiVersion: policy/v1beta1  
kind: PodSecurityPolicy  
metadata:  
  name: example-psp  
spec:  
  privileged: true
```